

Ex.no:5b	DATA VISUALIZATION USING MATPLOTLIB – SALES DATA

AIM

To visualize and analyze the **Life Expectancy dataset** using Line Chart, Bar Chart, and Pie Chart with Matplotlib to understand life expectancy trends over years, comparisons between developed and developing countries, and status distribution.

ALGORITHM

Step 1: Import Pandas and Matplotlib libraries.

Step 2: Load the Life Expectancy dataset into a DataFrame.

Step 3: Group the dataset according to year and calculate the average life expectancy for each year.

Step 4: Plot the yearly average life expectancy using a Line Chart to observe the trend over time.

Step 5: Group the dataset according to country status and calculate the average life expectancy for each status.

Step 6: Plot the average life expectancy using a Bar Chart to compare developed and developing countries.

Step 7: Select the country status values and calculate their proportions in the dataset.

Step 8: Plot the status distribution using a Pie Chart.

Step 9: Add suitable titles, axis labels, legends, and display the visualizations.

CODE

```
import pandas as pd
import os
import matplotlib.pyplot as plt

# Load the Life Expectancy dataset

path = "/content"

print("Dataset path:", path)

print("\nFiles in dataset:")
```

```

print(os.listdir(path))

# Find CSV file automatically

csv_files = [f for f in os.listdir(path) if f.lower().endswith(".csv")]

if not csv_files:
    raise FileNotFoundError("No CSV file found in the dataset path.")

csv_file = csv_files[0]

print("\nCSV file:", csv_file)

# Load dataset

df = pd.read_csv(
    os.path.join(path, csv_file),
    encoding="latin1",
    low_memory=False
)

print("\nDataset loaded successfully.")

```

```

# -----
# 1. Line Chart – Life Expectancy by Year
# -----

```

```

year_life = (
    df.groupby("Year")["Life expectancy"]
    .mean()
    .sort_index()
)

plt.figure(figsize=(10, 5))

plt.plot(
    year_life.index,
    year_life.values,
    marker="o"
)

plt.title("Average Life Expectancy by Year")

plt.xlabel("Year")

plt.ylabel("Average Life Expectancy")

plt.grid(True)

plt.show()

```

```

# -----
# 2. Bar Chart – Life Expectancy by Status
# -----

```

```

status_life = (
    df.groupby("Status")["Life expectancy"]
    .mean()
)

```

```

)

plt.figure(figsize=(8, 5))

plt.bar(
    status_life.index,
    status_life.values
)

plt.title("Average Life Expectancy by Country Status")

plt.xlabel("Country Status")

plt.ylabel("Average Life Expectancy")

plt.show()

```

```

# -----
# 3. Pie Chart – Country Status Distribution
# -----

```

```

status_counts = df["Status"].value_counts()

plt.figure(figsize=(8, 8))

plt.pie(
    status_counts.values,
    labels=status_counts.index,
    autopct="%1.1f%%",
    startangle=90
)

plt.title("Country Status Distribution")

plt.legend(
    status_counts.index,
    title="Status"
)

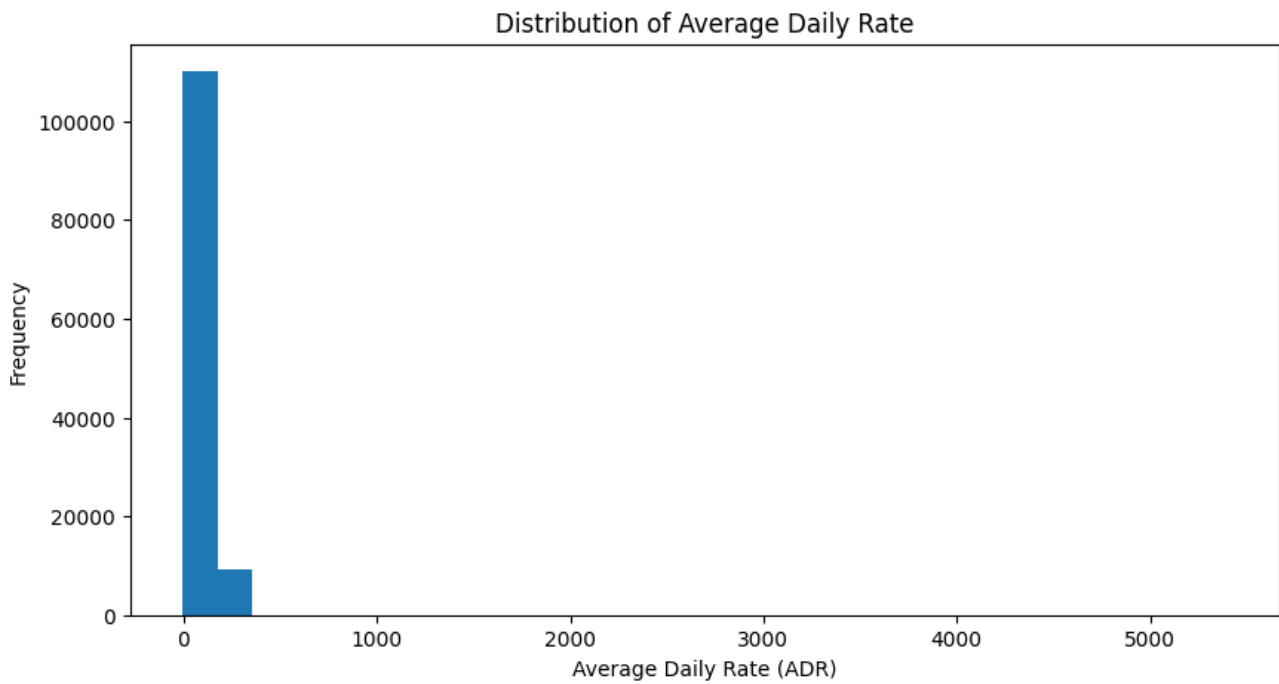
plt.show()

```

OUTPUT

Using Colab cache for faster access to the 'Life expectancy' dataset.

Mean : 69.22493169398908



Median : 72.1

Mode :

0 73.0

Name: Life expectancy, dtype: float64

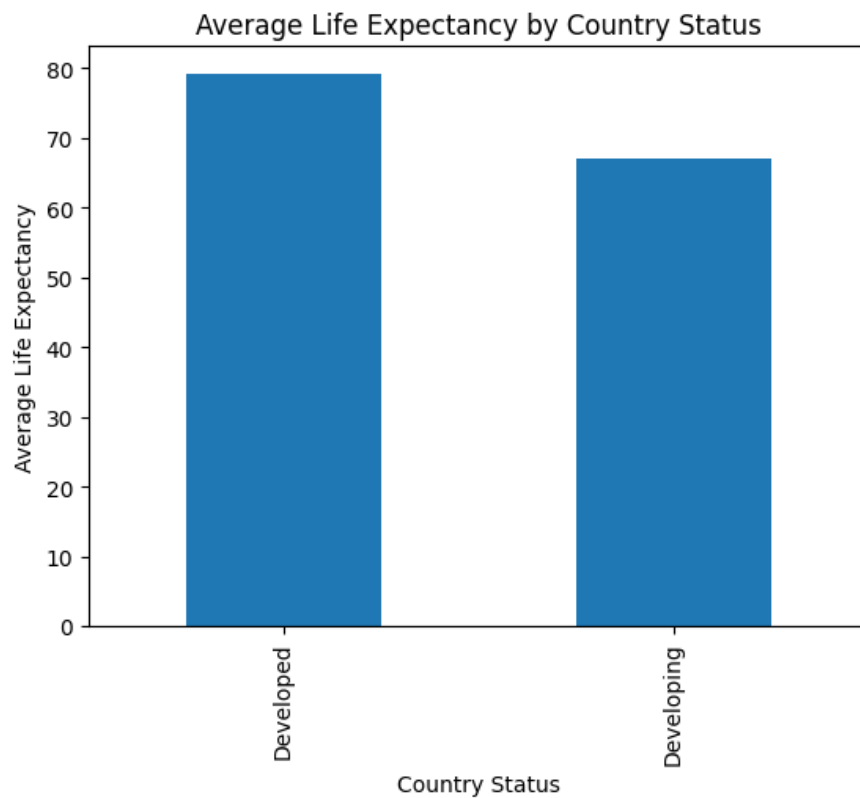
Standard Deviation : 9.523867487824278

Status

Developed 79.197852

Developing 67.111465

Name: Life expectancy, dtype: float64

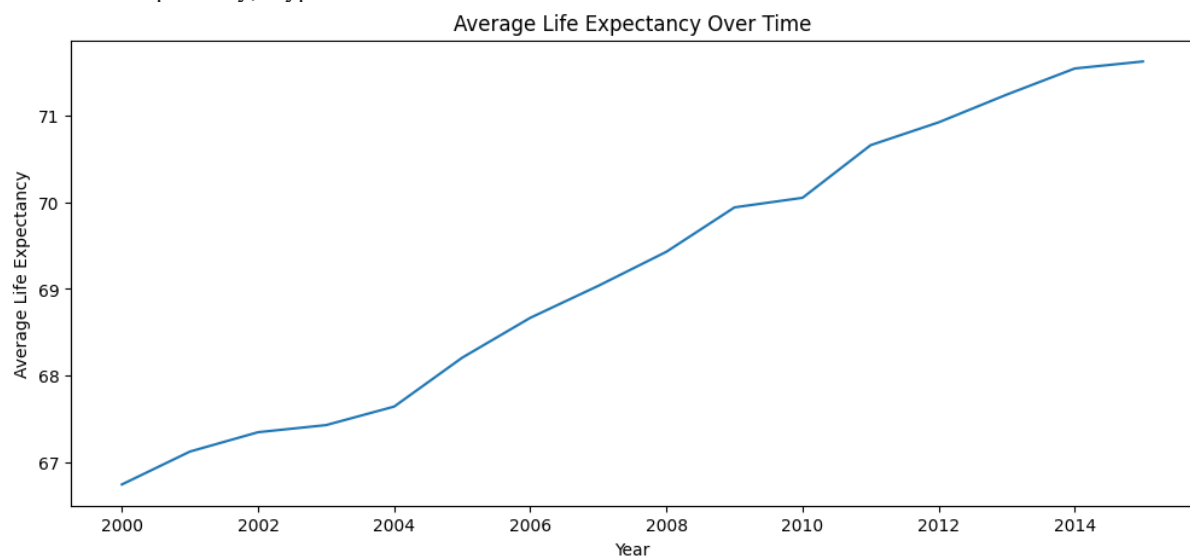


```

Year
2000  66.750273
2001  67.128962
2002  67.351366
2003  67.433333
2004  67.646448
2005  68.209290
2006  68.667760
2007  69.036066
2008  69.427869
2009  69.938251
2010  70.048634
2011  70.654098
2012  70.916940
2013  71.236066
2014  71.536612
2015  71.616940

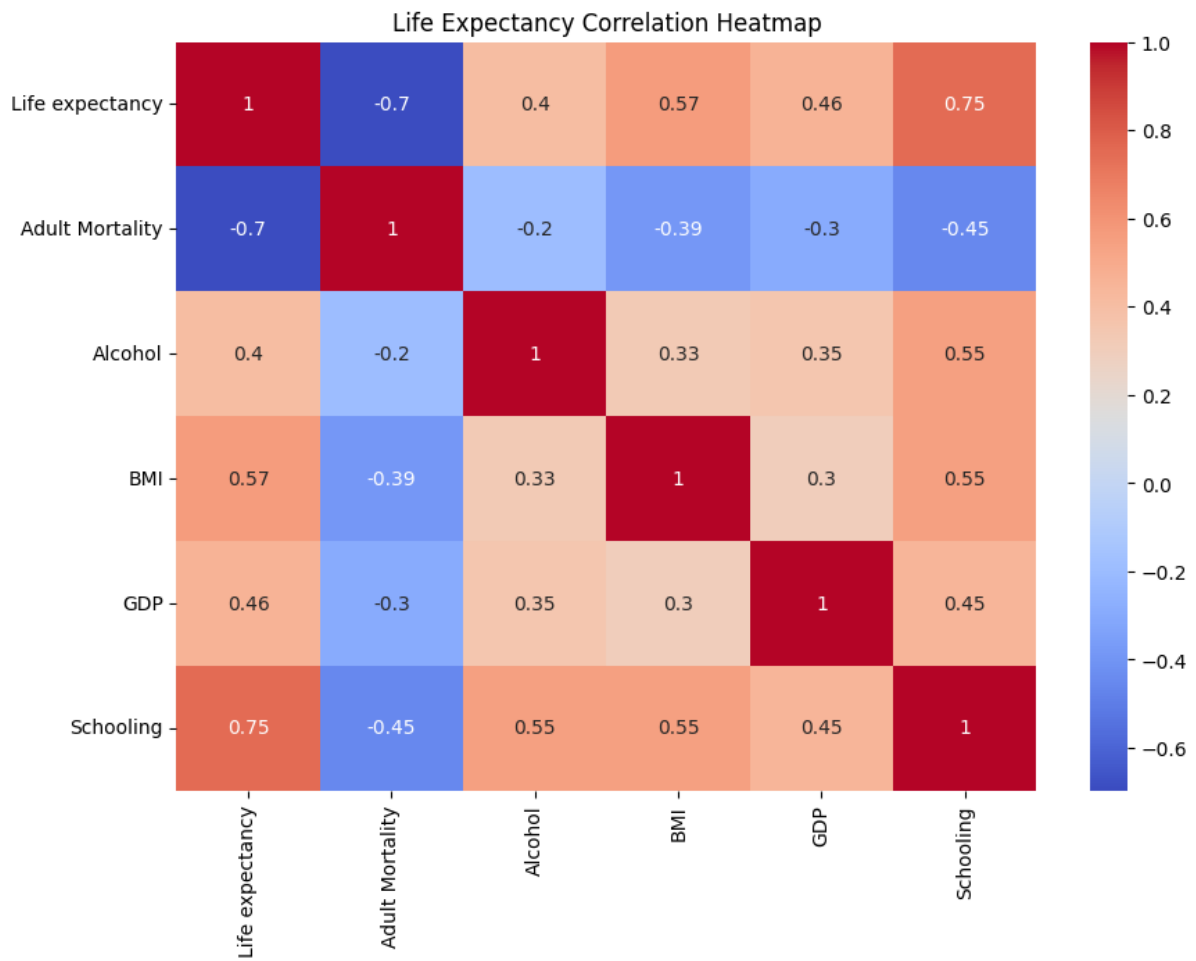
```

Name: Life expectancy, dtype: float64



	Life expectancy	Adult Mortality	Alcohol	BMI \
Life expectancy	1.000000	-0.696359	0.404877	0.567694
Adult Mortality	-0.696359	1.000000	-0.195848	-0.387017
Alcohol	0.404877	-0.195848	1.000000	0.330408
BMI	0.567694	-0.387017	0.330408	1.000000
GDP	0.461455	-0.296049	0.354712	0.301557
Schooling	0.751975	-0.454612	0.547378	0.546961

	GDP	Schooling
Life expectancy	0.461455	0.751975
Adult Mortality	-0.296049	-0.454612
Alcohol	0.354712	0.547378
BMI	0.301557	0.546961
GDP	1.000000	0.448273
Schooling	0.448273	1.000000



RESULT